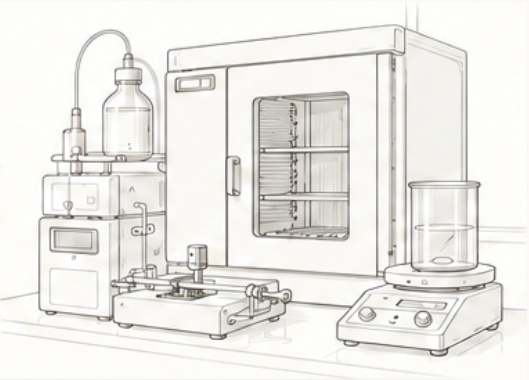
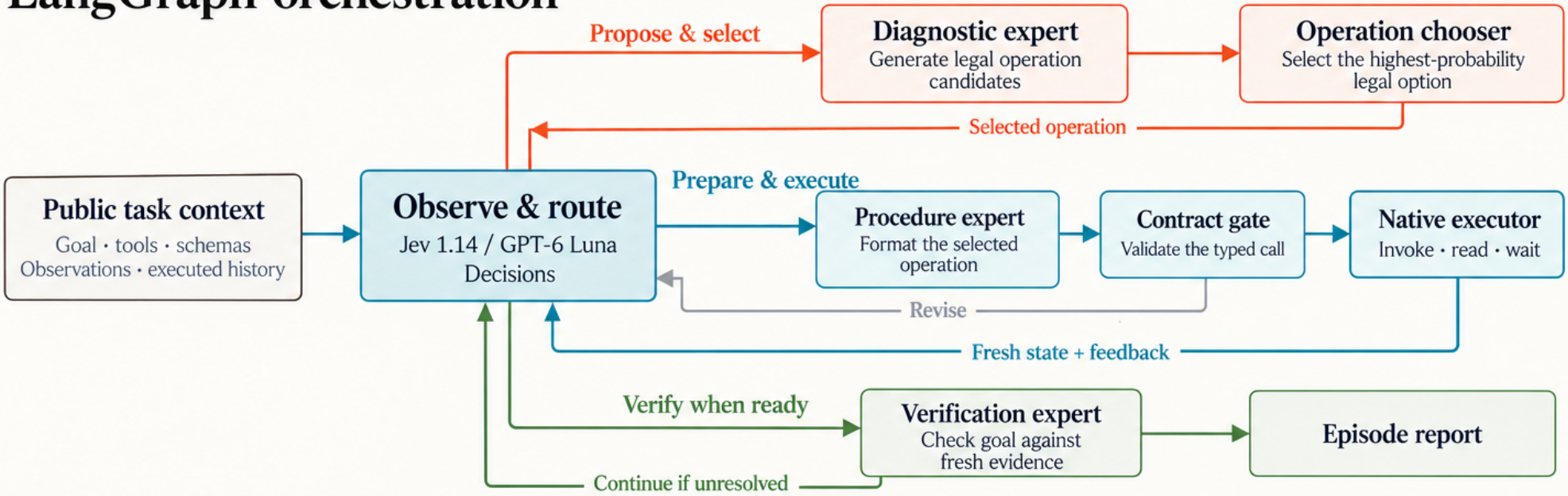
A decision-driven laboratory agent

Expert proposals. Explicit choices. Native execution.



LangGraph orchestration



Expert roles: MiniMax M3.1 Flash / DeepSeek V4.1 Flash

Routing + operation choice: Jev 1.14 / GPT-6 Luna Decisions

Contract-grounded actions

Public capabilities guide tool calls

Explicit decision layer

Expert reasoning is separated from selection

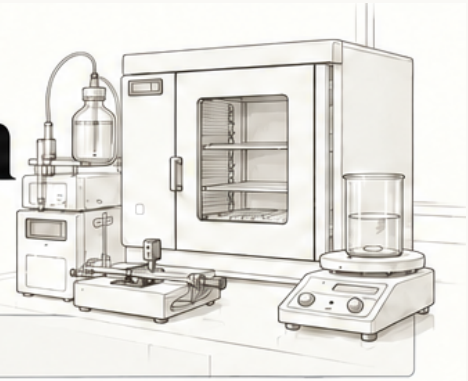
Closed-loop execution

Native observations drive the next step

Independent evaluator scores task success

From instrument knowledge to execution

Two complementary tracks for laboratory agents



Public instrument contracts • realistic operating scenarios • independent scoring

Knowledge & decision assessment

388 MCQA questions | **114** held-out

Five task categories

L1	Single-action checks	158
L2	Diagnosis & simple recovery	50
L3	Evidence, testing & escalation	88
L4	Procedure & service plan selection	62
L5	Interruption response selection	30

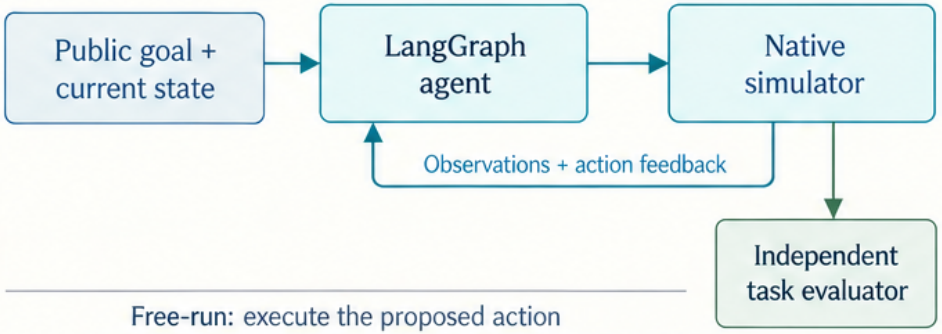
Question + supplied options → Direct or expert-assisted choice → Reference scoring

Hosted models • compact models • frozen features

Accuracy • category breakdown • held-out performance

Stateful workflow execution

120 stateful tasks | **14** workflows | **8** instrument families



Free-run: execute the proposed action
Teacher-forced: score, correct, then execute

6 free-run + 22 teacher-forced configurations

Autonomous success • original-action accuracy • assisted completion

One benchmark suite

Knowledge assessment and stateful execution.

Free-run

Measures autonomous end-to-end task completion.

Teacher-forced

Measures original decisions along corrected trajectories.

L1–L5 denote task categories.

Instrument Agent Evaluation

KNOWLEDGE AND EXECUTION · MCQA AND AUTONOMOUS WORKFLOWS

Knowledge: MiniMax + Jev reaches 96.9% across 388 questions.
Execution: MiniMax max + Jev completes 58.3% of tasks; MiniMax max + Luna reaches 54.2%.

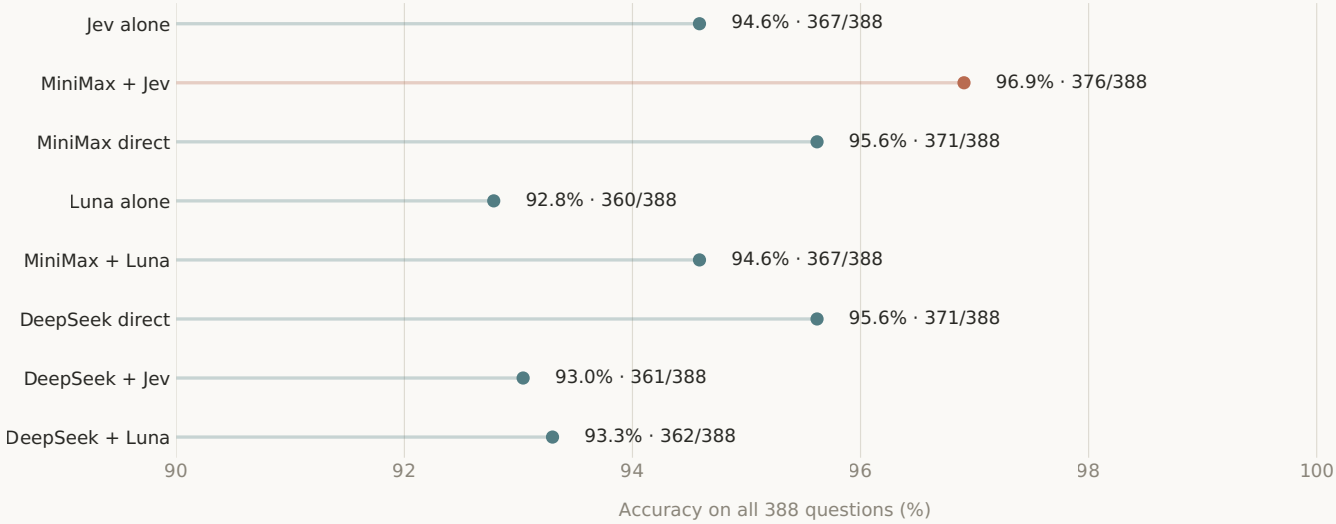
Models and versions. Hosted models are accessed through OpenRouter. Expert models: MiniMax M3.1 Flash (`minimax/minimax-m3.1-flash`) and DeepSeek V4.1 Flash (`deepseek/deepseek-v4.1-flash`). Decision models: TypeSafe Jev 1.14 (`typesafe/jev-1.14`) and OpenAI GPT-6 Luna Decisions (`openai/gpt-6-luna-decisions`). MiniMax workflow runs compare low and max reasoning; DeepSeek workflow runs use low.

01 / INSTRUMENT KNOWLEDGE

Multiple-choice question answering

Eight hosted-model configurations answer the same 388 questions. Accuracy measures selection of the accepted answer from the supplied options.

Best overall: MiniMax + Jev, 376/388 correct (96.9%).



METHOD	CORRECT / 388	ACCURACY
Jev alone	367/388	94.6%
MiniMax + Jev	376/388	96.9%
MiniMax direct	371/388	95.6%
Luna alone	360/388	92.8%
MiniMax + Luna	367/388	94.6%
DeepSeek direct	371/388	95.6%
DeepSeek + Jev	361/388	93.0%
DeepSeek + Luna	362/388	93.3%

Performance by task category

Five categories describe the type of decision being tested.

CATEGORY	DECISION BEING TESTED	QUESTIONS
L1	Single-action checks	158
L2	Direct diagnosis and simple recovery	50
L3	Evidence assessment, testing and escalation	88
L4	Procedure and service plan selection	62
L5	Interruption response plan selection	30

	L1 · n=158	L2 · n=50	L3 · n=88	L4 · n=62	L5 · n=30
Jev alone	96.8%	100.0%	96.6%	100.0%	56.7%
MiniMax + Jev	100.0%	98.0%	89.8%	98.4%	96.7%
MiniMax direct	100.0%	90.0%	87.5%	100.0%	96.7%
Luna alone	96.2%	92.0%	83.0%	98.4%	93.3%
MiniMax + Luna	98.1%	96.0%	88.6%	93.5%	93.3%
DeepSeek direct	99.4%	100.0%	84.1%	100.0%	93.3%
DeepSeek + Jev	98.1%	98.0%	79.5%	98.4%	86.7%
DeepSeek + Luna	97.5%	98.0%	78.4%	100.0%	93.3%

Jev excels at evidence assessment, but struggles with interruption response. Jev alone leads L3 at 85/88 (96.6%), while answering only 17/30 L5 questions correctly (56.7%).

MiniMax advice shifts the balance. MiniMax + Jev raises L5 to 29/30 (96.7%), while L3 falls to 79/88 (89.8%). The stronger overall score reflects different gains and losses across categories.

Basic checks and supplied procedures are near ceiling. L1 scores range from 96.2–100% and L4 from 93.5–100%. L1 contributes 158/388 questions (40.7%) to the overall score. L4 and L5 assess selection of supplied plans.

Smaller models, different decision mechanisms

The knowledge evaluation adds 34 local configurations and references, spanning native decision heads, generative choices, frozen-feature readouts and expert–chooser combinations. These methods are compared on 114 held-out questions; training and validation examples are excluded from this score.



APUS Reflex high leads the compact models. It answers 106/114 held-out questions correctly (93.0%). APUS low scores 71.1%, while greedy Qwen3 4B scores 78.1% and Qwen2.5 3B scores 64.9%.

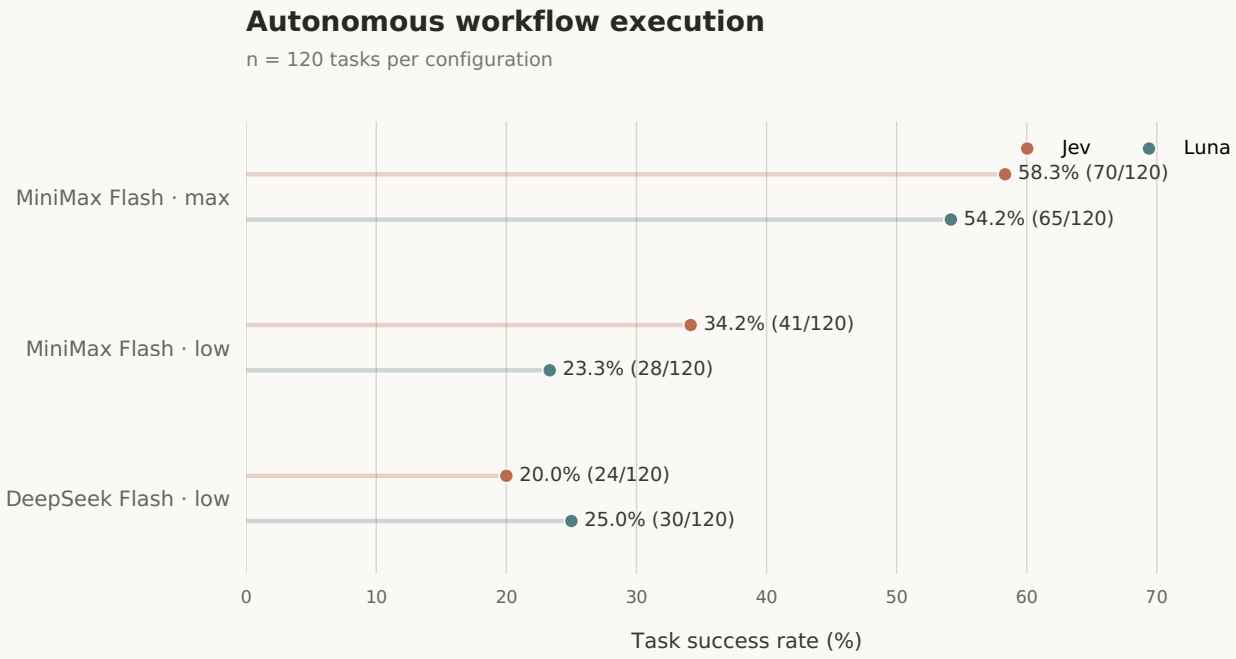
Adding an expert does not automatically improve selection. Qwen2.5 advice followed by APUS high achieves 67.5%, versus 93.0% for the native APUS high head. Frozen-feature methods span 62.3–70.2%; their readouts use the MCQA training split, and cosine temperature uses validation data.

The chart shows native heads, HF greedy choices, four feature methods, all four expert–chooser combinations and two references. Every displayed method uses 114 held-out questions; the separate T1 rule reference covers 64. OpenJev-8B is a separate checkpoint from hosted Jev 1.14.

Local checkpoints. Qwen2.5-3B-Instruct; Qwen3-4B-Instruct-2507; OpenJev-8B; APUS Reflex 4B-5949 (low/high); all-MiniLM-L6-v2.

Autonomous instrument workflows

120 tasks across 14 workflows and 8 instrument families. Six expert–decision model combinations execute tasks through the LangGraph workflow.



Results at a glance

MiniMax with stronger reasoning leads this evaluation. Jev performs better with MiniMax low; Luna performs better with DeepSeek low. The decision-model advantage depends on the expert configuration.

EXPERT / REASONING	DECISION MODEL	SUCCESS	RATE	OUTCOME MIX	NATIVE UNSUCCESSFUL	EARLY EXIT
MiniMax · max	Jev	70/120	58.3%		19	31
MiniMax · max	Luna	65/120	54.2%		14	41
MiniMax · low	Jev	41/120	34.2%		8	71
MiniMax · low	Luna	28/120	23.3%		7	85
DeepSeek · low	Jev	24/120	20.0%		2	94
DeepSeek · low	Luna	30/120	25.0%		6	84

Green: success · Amber: native unsuccessful · Rose: early exit.

What changes the outcome?

Reasoning intensity

With Luna, MiniMax rises from 28 to 65 successes: +37 tasks, or +30.8 percentage points. With Jev, max achieves 70 versus 41 at low.

Decision model

At MiniMax low, Jev leads Luna by 13 tasks (+10.8 pp). At DeepSeek low, Luna leads Jev by 6 tasks (+5.0 pp).

Failure pattern

Many unsuccessful runs end before normal evaluator completion. The table below separates observable output/call errors from external failures and unresolved causes.

Instrument coverage

INSTRUMENT	MINIMAX · MAX JEV	MINIMAX · MAX LUNA	MINIMAX · LOW JEV	MINIMAX · LOW LUNA	DEEPSEEK · LOW JEV	DEEPSEEK · LOW LUNA
binder_fd056	9/18 · 50%	10/18 · 56%	6/18 · 33%	4/18 · 22%	7/18 · 39%	7/18 · 39%
eppendorf_cellxpertc170i	1/6 · 17%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%
esco_efa5uduvw	11/18 · 61%	12/18 · 67%	10/18 · 56%	7/18 · 39%	2/18 · 11%	1/18 · 6%
ika_plate_rctdigital	16/18 · 89%	15/18 · 83%	8/18 · 44%	7/18 · 39%	4/18 · 22%	6/18 · 33%
nabertherm_l9	15/18 · 83%	11/18 · 61%	8/18 · 44%	6/18 · 33%	7/18 · 39%	11/18 · 61%
newera_ne1000	16/24 · 67%	13/24 · 54%	6/24 · 25%	3/24 · 12%	3/24 · 12%	4/24 · 17%
peak_precisionhydrogen	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%
vaisala_dmt152	2/12 · 17%	4/12 · 33%	3/12 · 25%	1/12 · 8%	1/12 · 8%	1/12 · 8%

Workflow performance

WORKFLOW	MINIMAX · MAX JEV	MINIMAX · MAX LUNA	MINIMAX · LOW JEV	MINIMAX · LOW LUNA	DEEPSEEK · LOW JEV	DEEPSEEK · LOW LUNA
dewpoint_offline_calibration	2/6 · 33%	4/6 · 67%	3/6 · 50%	1/6 · 17%	1/6 · 17%	1/6 · 17%
dewpoint_startup_measurement_diagnosis	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%
furnace_door_pause_recovery	5/6 · 83%	3/6 · 50%	2/6 · 33%	1/6 · 17%	4/6 · 67%	5/6 · 83%
furnace_program	10/12 · 83%	8/12 · 67%	6/12 · 50%	5/12 · 42%	3/12 · 25%	6/12 · 50%
hood_illumination	10/12 · 83%	8/12 · 67%	7/12 · 58%	7/12 · 58%	1/12 · 8%	0/12 · 0%
hood_sash_access_resecure	1/6 · 17%	4/6 · 67%	3/6 · 50%	0/6 · 0%	1/6 · 17%	1/6 · 17%
hotplate_motor_diagnostic	6/6 · 100%	4/6 · 67%	4/6 · 67%	1/6 · 17%	1/6 · 17%	2/6 · 33%
hotplate_unlock_stirring	10/12 · 83%	11/12 · 92%	4/12 · 33%	6/12 · 50%	3/12 · 25%	4/12 · 33%
hydrogen_startup_diagnosis	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%
incubator_sensor_cover_access	1/6 · 17%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%
oven_standby_restoration	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%	0/6 · 0%
oven_usb_recording	9/12 · 75%	10/12 · 83%	6/12 · 50%	4/12 · 33%	7/12 · 58%	7/12 · 58%
pump_delivery	6/12 · 50%	5/12 · 42%	1/12 · 8%	0/12 · 0%	0/12 · 0%	0/12 · 0%
pump_timed_monitoring	10/12 · 83%	8/12 · 67%	5/12 · 42%	3/12 · 25%	3/12 · 25%	4/12 · 33%

Early-exit attribution

Categories describe the terminal evidence. Model output/call errors include malformed JSON and calls outside the supplied contract. Unresolved cases remain unassigned rather than being attributed to the decision model.

CONFIGURATION	MODEL OUTPUT / CALL	EXTERNAL SERVICE / RUNTIME	UNRESOLVED
MiniMax · max + Jev	30	0	1
MiniMax · max + Luna	38	0	3
MiniMax · low + Jev	69	0	2
MiniMax · low + Luna	85	0	0
DeepSeek · low + Jev	77	13	4
DeepSeek · low + Luna	71	11	2

Token usage

Recorded input, output, and reasoning tokens across each configuration. Reasoning tokens are included in output totals.

CONFIGURATION	INPUT M	OUTPUT M	REASONING M (WITHIN OUTPUT)
MiniMax · max + Jev	45.214	4.005	3.500
MiniMax · max + Luna	38.622	3.924	3.491
MiniMax · low + Jev	34.804	0.668	0.244
MiniMax · low + Luna	27.590	0.528	0.200
DeepSeek · low + Jev	35.474	2.471	1.750
DeepSeek · low + Luna	32.492	2.361	1.764

Two execution modes, complementary measurements

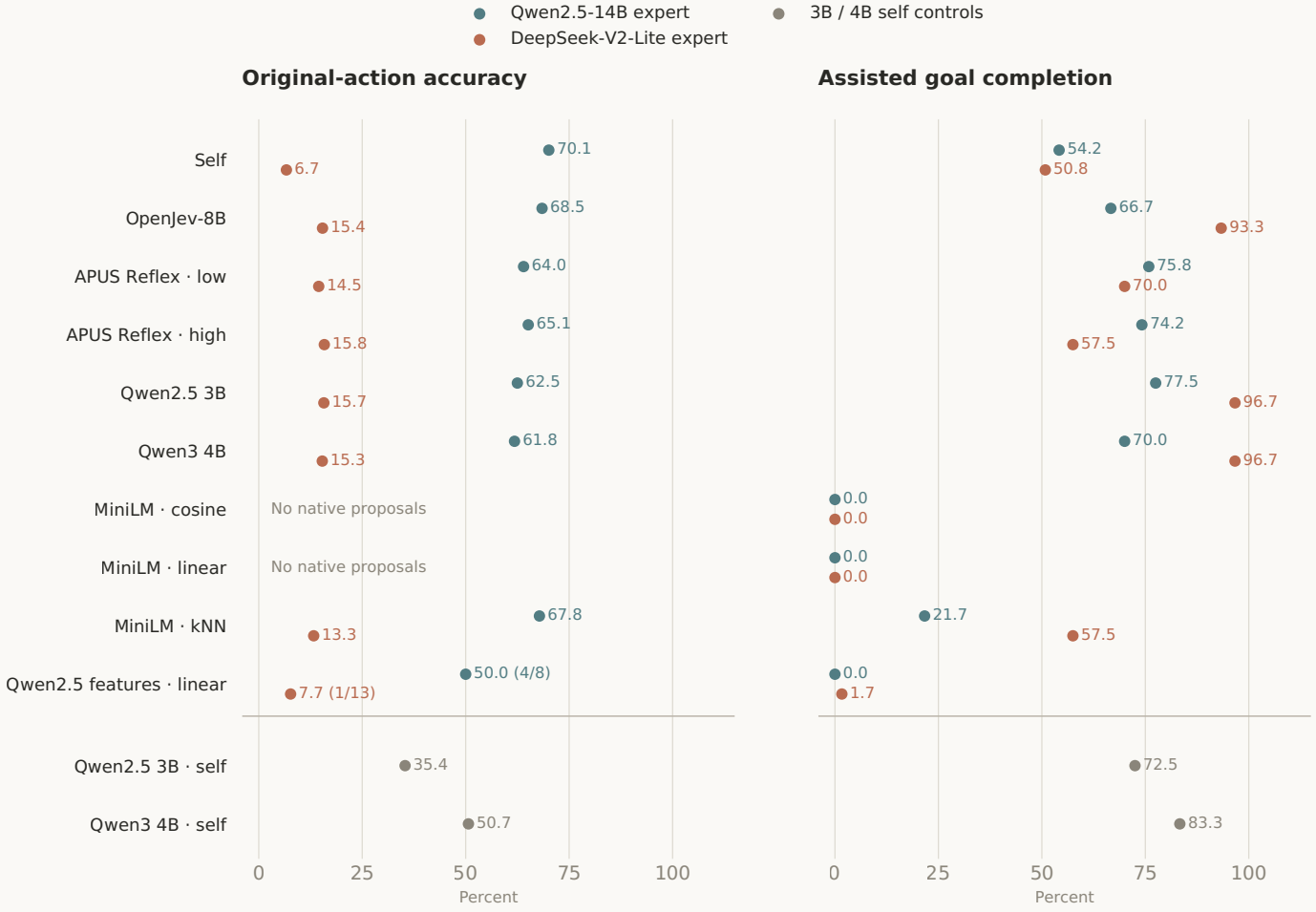
	FREE-RUN	TEACHER-FORCED
Execution	The agent's proposed action is attempted; subsequent choices follow the resulting state.	The original action is scored; an incorrect action is replaced by an accepted reference action.
What it measures	Autonomous task completion across the full trajectory.	Original-action accuracy and assisted completion along corrected trajectories.
Reported configurations	6 hosted expert–decision combinations	22 local expert–chooser combinations

Free-run captures error propagation and recovery. Teacher-forced keeps later decisions assessable after an incorrect proposal, including for models operating with compact context views. Both modes use the same 120-task benchmark; their completion rates answer different questions.

Decision quality along corrected trajectories

Twenty-two local expert–chooser configurations each attempt all 120 tasks. The benchmark grades every original action before applying a reference correction, allowing later decisions to be evaluated along a corrected trajectory. Native feedback returns to the same LangGraph workflow.

Original-action accuracy: 5,477/12,466 (43.9%).
Assisted goal completion: 1,344/2,640 (50.9%), with 6,989 action corrections.



Completion and independent decision quality diverge. DeepSeek-V2-Lite with Qwen2.5 3B or Qwen3 4B reaches 116/120 assisted completions (96.7%), while original-action accuracy is only 15.7% and 15.3%. Qwen2.5-14B selecting for itself reaches 70.1% original-action accuracy, but completes 65/120 assisted episodes (54.2%).

Reaching a native action remains a challenge. Four MiniLM cosine/linear combinations produce no native proposals. Their original-action accuracy is unavailable, and assisted completion is zero; corrections can begin only after the graph emits a native action.

Action counts include observed proposals before saved aborts; each completion denominator is 120. The 2,640 records comprise 1,344 completed episodes and 1,296 model-output or budget aborts. No accuracy is assigned when no native proposal is observed.

Local experts are Qwen2.5-14B-Instruct and DeepSeek-V2-Lite-Chat, plus the 3B/4B self controls. These runs use bounded current-decision views, schema-guided outputs and a deterministic procedure envelope. Hosted free-run runs use a generative procedure role.